

HYDROGENATED NBR (HNBR) SYNTHESIS AND PROCESS CONTROL ACROSS CATALYST-DRIVEN HYDROGENATION PATENTS

1. ABSTRACT

This report synthesizes five patents focused on hydrogenated nitrile rubber (HNBR), with emphasis on selective hydrogenation of nitrile-butadiene rubber (NBR) precursors to reduce residual unsaturation while retaining polymer integrity and processability. The target material is consistently defined as HNBR derived from at least one conjugated diene and at least one α,β -unsaturated nitrile, typically acrylonitrile-butadiene rubber, with desired properties including improved heat, oil, and ozone resistance and low residual double bond (RDB) content. The selected patent landscape spans two major synthesis strategies: conventional solution hydrogenation of NBR using noble-metal catalysts, and simultaneous metathesis/hydrogenation routes that reduce molecular weight and narrow molecular-weight distribution in a single step. Across the patents, ruthenium-, osmium-, rhodium-, palladium-, and Wilkinson-type catalyst systems are disclosed, with strong emphasis on catalyst composition, catalyst loading, hydrogen pressure, temperature, reaction time, and co-catalyst or co-olefin effects.

US10508156B2 discloses catalyst compositions based on ruthenium or osmium complexes pretreated with vinyl ether co-catalysts to improve hydrogenation activity and preserve molecular weight during HNBR formation. US7772328B2 and EP2473280B1 both address low-molecular-weight HNBR via simultaneous metathesis and hydrogenation, using Ru/Os metathesis catalysts and optional co-olefins to cleave polymer chains while saturating residual double bonds. EP2473280B1 compares Grubbs III, Grubbs II, and Wilkinson's catalyst and shows faster hydrogenation with Grubbs III at equal metal content. US9150669B2 extends this concept with alternative Ru/Os catalyst structures and demonstrates that catalyst choice strongly affects hydrogenation rate, RDB reduction, and molecular-weight retention. EP3827029A1 focuses less on the hydrogenation chemistry itself and more on production quality, catalyst residue control, and aging stability, specifying noble-metal loading windows and reaction conditions for producing HNBR with improved aging properties.

Overall, the evidence supports HNBR synthesis routes that combine selective hydrogenation with catalyst systems engineered to control metathesis, gel formation, catalyst residue, and molecular-weight loss. The most robust process window in the evidence uses solution-phase hydrogenation in organic solvent, hydrogen pressures around 4.137–8.4 MPa, temperatures from 120–140°C in the examples, and catalyst/co-catalyst pretreatment to tune activity and product

properties.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US10508156B2
- Patent Title: Catalyst compositions and their use for hydrogenation of nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses selective hydrogenation of nitrile rubber using catalyst compositions formed by contacting a ruthenium- or osmium-based carbene complex with a vinyl-containing co-catalyst, especially vinyl ethyl ether, before hydrogenation. The catalyst pretreatment is used either separately or in situ to control metathesis activity while improving hydrogenation performance and retaining polymer molecular weight. - Catalyst to co-catalyst molar ratio: 1:(20 to below 100) — Claims - Hydrogenation temperature: 120° C. — Table 2 / Examples 1-5 - Hydrogen pressure: 4.137 MPa — Table 2 / Examples 1-5 - Agitation speed: 800 rpm — Example 1-5 - Perbunan® 3431VP concentration: 6 wt % — Example 1-5 - Perbunan® 3431VP amount: 18 g — Example 1-5 - MCB amount: 282 g + 22 g — Example 1-5 - Wilkinson's catalyst amount: 15 mg — Example 1 - PPh₃ amount: 18 mg — Example 1 - Catalyst (1) amount: 9 mg — Example 2 - Vinyl ethyl ether amount: 100 µL — Example 2 - Catalyst pretreatment time: 12 hours — Example 2 - Catalyst (2) amount: 18 mg — Example 3 - Catalyst (2) amount: 9 mg — Example 4 - Catalyst (3) amount: 9 mg — Example 5 - NBR-5 amount: 270 g — Example 6 - MCB amount: 4,350 g — Example 6 - 1-hexene amount: 11.1 g — Example 6 - Catalyst (2) amount: 135 mg — Example 6 - Hydrogenation temperature: 140° C. — Example 6 - Hydrogen pressure: 8.4 MPa — Example 6 - Agitation speed: 600 rpm — Example 6 - NBR-6 amount: 270 g — Example 7 - MCB amount: 4,350 g — Example 7 - Catalyst (2) amount: 135 mg — Example 7 - Vinyl ethyl ether amount: 0.75 g — Example 7 - Catalyst pretreatment time: 12 hours — Example 7 - 1-hexene amount: 11.1 g — Example 7 - NBR-6 amount: 560 g — Examples 8-10 - NBR-6 concentration: 11 wt % — Examples 8-10 - 1-hexene amount: 23.1 g — Examples 8-10 - Catalyst (2) amount: 280 mg — Examples 8-10 - Catalyst metathesis time: 15 min — Examples 8-10 - Vinyl ethyl ether amount: 1.55 g — Examples 8-10 - Hydrogenation temperature: 140° C. — Table 3 / Examples 6-10 - Hydrogenation pressure: 8.4 MPa — Table 3 / Examples 6-10 - Hydrogenation time: 2 h — Table 3 / Examples 6-10 - NBR-7 amount: 518 g — Examples 11-12 - NBR-7 concentration: 13 wt % — Examples 11-12 - Catalyst

(2) amount: 363 mg — Example 11 - Catalyst (2) amount: 259 mg — Example 12 - Vinyl ethyl ether amount: 1.41 g — Example 12 - Catalyst pretreatment time: 1 hour — Example 12 - Hydrogenation pressure: 8.4 MPa — Example 11-12 - Hydrogenation temperature: 140° C. — Example 11-12 - Agitation speed: 600 rpm — Example 11-12

Relevant Experimental Evidence - Example 1: Comparative hydrogenation with Wilkinson's catalyst reached 90.3% hydrogenation in 5 hours and retained Mn 76,286 g/mol and Mw 260,572 g/mol. - Example 2: Catalyst (1) pretreated with vinyl ethyl ether gave 93% hydrogenation in 3 hours with lower PDI (2.95) than Example 1. - Example 3: Catalyst (2) with vinyl ethyl ether achieved 99% hydrogenation in 1 hour. - Example 6: A metathesis/hydrogenation sequence on NBR-5 gave 99.5% hydrogenation in 2 hours but with strong molecular-weight reduction. - Example 12: Catalyst (2) pretreated with vinyl ethyl ether achieved >99% hydrogenation using less catalyst than the non-pretreated comparison example.

Technical Relevance This patent is highly relevant because it directly addresses HNBR synthesis by selective hydrogenation of nitrile rubber and shows how catalyst pretreatment with vinyl ethers can improve hydrogenation efficiency while controlling molecular-weight retention. It also provides practical process windows for pressure, temperature, and catalyst loading in solution hydrogenation.

Patent 2

- Patent Number: US7772328B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses production of low-molecular-weight hydrogenated nitrile rubber by first reducing the molecular weight of nitrile rubber through olefin metathesis with a co-olefin and then hydrogenating the degraded polymer. The process is carried out in solution using ruthenium- or osmium-based metathesis catalysts and is intended to deliver HNBR with lower Mooney viscosity and narrower molecular-weight distribution than conventional direct hydrogenation. - Cement concentration: 15 wt.-% solid — Example 1 - Cement concentration: 15 wt.-% solid — Example 2 - Cement concentration: 6 wt.-% solid — Example 3 - Cement concentration: 6 wt.-% solid — Example 4 - C₂H₄ pressure: 6.9×10⁵ Pa — Examples 1-3 - 1-hexene amount: 150 g — Example 4 - Dry N₂ degassing: 3 times — Example 4 - Co-olefin pressure: 500 psi — Table 1 - Reactor temperature: 80° C. — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Mn: 27 kg/mol — text summary - Mw: 54 kg/mol — text summary - PDI: 2.0 — text summary - Mw reduction: up to fivefold — Table 2

summary - PDI minimum: 1.90 — Table 2 summary - Mooney viscosity: 73 — Therban® A3407 substrate - Mooney viscosity: 43 — Experiment 1 - Mooney viscosity: 28 — Experiment 2 - Mooney viscosity: 3 — Experiment 3 - Mooney viscosity: 31 — Experiment 4 - Mn: 98000 — Therban® A3407 substrate - Mw: 320000 — Therban® A3407 substrate - MZ: 945000 — Therban® A3407 substrate - PDI: 3.27 — Therban® A3407 substrate

Relevant Experimental Evidence - Example 1: 200 g rubber in MCB was degassed with ethylene at 6.9×10^5 Pa before metathesis/hydrogenation. - Example 2: Same 15 wt-% cement and ethylene degassing conditions were repeated as a comparative run. - Example 3: Lower-solids cement (6 wt-%) was used under the same ethylene degassing conditions. - Example 4: 1-hexene was used as co-olefin after charging 75 g rubber in 1175 g MCB. - Example 4: Product properties showed reduced Mw and narrowed PDI relative to the starting polymer.

Technical Relevance This patent is relevant because it demonstrates a route to HNBR with deliberately reduced molecular weight and narrowed distribution, which is important for improving processability. It also provides explicit evidence that co-olefin-assisted metathesis followed by hydrogenation can generate low-viscosity HNBR grades.

Patent 3

- Patent Number: EP3827029A1
- Patent Title: Method for producing hydrogenated nitrile rubber and hnbr compositions thereof
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a method for producing HNBR in solution using ruthenium, palladium, or rhodium compounds while emphasizing catalyst residue control and improved aging properties. The process is framed around hydrogenating partially unsaturated nitrile rubber under defined noble-metal loading, temperature, pressure, and time windows, with downstream removal/recovery of catalyst residues as a key quality objective. - Ruthenium amount: 10 ppm to 200 ppm — Claim 1 - Ruthenium amount: 10 ppm to 150 ppm — Claim 1 - Ruthenium amount: 10 ppm to 120 ppm — Claim 1 - Ruthenium amount: 10 ppm to 79 ppm — Claim 1 - Ruthenium amount: 42 ppm to 79 ppm — Claim 1 - Palladium amount: 20 ppm to <67 ppm — Claim 1 - Palladium amount: 44 ppm to <67 ppm — Claim 1 - Rhodium amount: 50 ppm to <270 ppm — Claim 1 - Rhodium amount: 79 ppm to <270 ppm — Claim 1 - Rhodium amount: 50 ppm to 250 ppm — Claim 1 - Rhodium amount: 50 ppm to 170 ppm — Claim 1 - Rhodium amount: 79 ppm to 170 ppm — Claim 1 - Hydrogenation temperature: 60 to 200°C — Claim 1 - Hydrogenation pressure: 700 000 pascals to 15 000 000 pascals — Claim 1 - Hydrogenation time: 1 to 24 hours — Claim 1 - RDB increase during ageing: less than 10% — Description - Ageing condition: 4 days at 140°C — Description - Mooney

viscosity increase during ageing: less than 40% — Description - Rhodium content: 58-60 ppm — prior art cited in description - Rhodium content: 5.8 to 9.2 ppm — prior art cited in description - Pd catalyst loading: 0.45 parts — prior art cited in description

Relevant Experimental Evidence - No experimental evidence extracted from available evidence.

Technical Relevance This patent is relevant because it defines a practical HNBR production window centered on noble-metal-catalyzed hydrogenation and catalyst-residue management. Its main contribution is process quality control: minimizing catalyst residues to improve aging behavior, color, and gel stability in the final HNBR.

Patent 4

- Patent Number: EP2473280B1
- Patent Title: Process for the preparation of hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a one-step process in which nitrile rubber is simultaneously hydrogenated and metathesized in the presence of hydrogen, optionally a co-olefin, and a ruthenium- or osmium-based metathesis catalyst. The approach is designed to produce HNBR with lower molecular weight and narrower molecular-weight distribution than conventional hydrogenation routes. - Catalyst amount: 0,057 phr — Table 1 / HNBR 1 - Catalyst amount: 0,080 phr — Table 1 / HNBR 2 - Catalyst amount: 0,06 phr — Table 1 / HNBR 3 - Catalyst amount: 0,055 phr — Table 1 / HNBR 4 - Metal content: 65,6 ppm — Table 1 / HNBR 1, 3, 4 - Metal content: 91,3 ppm — Table 1 / HNBR 2 - Reaction time: 60 min — Table 2 - Reaction time: 120 min — Table 2 - Reaction time: 180 min — Table 2 - Reaction time: 240 min — Table 2 - RDB: 2,0% at 60 min — Table 2 / HNBR 1 - RDB: 0,7% at 60 min — Table 2 / HNBR 2 - RDB: 15,4% at 60 min — Table 2 / HNBR 3 - RDB: 2,9% at 60 min — Table 2 / HNBR 4 - RDB: 0,4% at 120 min — Table 2 / HNBR 1 - RDB: 0,5% at 120 min — Table 2 / HNBR 2 - RDB: 3,1% at 120 min — Table 2 / HNBR 3 - RDB: 0,6% at 120 min — Table 2 / HNBR 4 - RDB: 0,3% at 180 min — Table 2 / HNBR 1 - RDB: 0,3% at 180 min — Table 2 / HNBR 2 - RDB: 1,3% at 180 min — Table 2 / HNBR 3 - RDB: 0,3% at 180 min — Table 2 / HNBR 4 - Mw: 191541 g/mol — Table 3 / HNBR 1 - Mn: 68643 g/mol — Table 3 / HNBR 1 - PDI: 2,79 — Table 3 / HNBR 1 - Mooney viscosity: 38,2 — Table 3 / HNBR 1 - Mw: 135772 g/mol — Table 3 / HNBR 2 - Mn: 58259 g/mol — Table 3 / HNBR 2 - PDI: 2,33 — Table 3 / HNBR 2 - Mooney viscosity: 20,2 — Table 3 / HNBR 2 - Mw: 195304 g/mol — Table 3 / HNBR 3 - Mn: 64130 g/mol — Table 3 / HNBR 3 - PDI: 3,05 — Table 3 / HNBR 3 - Mooney viscosity: 69,3 — Table 3 / HNBR 3 - Mw: 144212 g/mol — Table 3 / HNBR 4 - Mn: 59851 g/mol — Table 3 / HNBR 4 - PDI: 2,45 — Table 3 / HNBR 4 - Mooney viscosity: 32,6 — Table 3 / HNBR 4

Relevant Experimental Evidence - Example 1: Grubbs III, Grubbs II, and Wilkinson's catalyst were compared at equal metal content. - Example 1: Grubbs III showed faster hydrogenation than Grubbs II or Wilkinson's catalyst over the first 120 min. - Example 1: HNBR 2 achieved the lowest Mooney viscosity among the compared samples (20,2). - Example 1: HNBR 1 and HNBR 4 showed low RDB values after 240 min. - Example 1: Product molecular weights remained in the low-to-mid 10^5 g/mol range.

Technical Relevance This patent is directly relevant because it combines metathesis and hydrogenation to make lower-molecular-weight HNBR in a single step. It also provides comparative catalyst data showing that catalyst identity strongly affects hydrogenation rate, residual unsaturation, and final viscosity.

Patent 5

- Patent Number: US9150669B2
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a one-step process for producing low-molecular-weight HNBR by contacting nitrile rubber with a ruthenium- or osmium-based metathesis catalyst in the presence of hydrogen and optionally a co-olefin. The process is intended to improve hydrogenation efficiency while simultaneously reducing molecular weight and narrowing molecular-weight distribution. - Catalyst amount: 0.041 phr — Table 1 / HNBR 1 - Catalyst amount: 0.045 phr — Table 1 / HNBR 2 - Catalyst amount: 0.060 phr — Table 1 / HNBR 3 - Metal content: 65.6 ppm — Table 1 / HNBR 1-3 - Co-catalyst: NIL — Table 1 / HNBR 1-2 - Co-catalyst: Triphenylphosphine — Table 1 / HNBR 3 - Co-catalyst amount: 1.0 phr — Table 1 / HNBR 3 - Reaction time: 60 min — Table 2 - Reaction time: 120 min — Table 2 - Reaction time: 180 min — Table 2 - Reaction time: 240 min — Table 2 - RDB: 2.5% at 60 min — Table 2 / HNBR 1 - RDB: 7.0% at 60 min — Table 2 / HNBR 2 - RDB: 15.4% at 60 min — Table 2 / HNBR 3 - RDB: 0.6% at 120 min — Table 2 / HNBR 1 - RDB: 4.5% at 120 min — Table 2 / HNBR 2 - RDB: 3.1% at 120 min — Table 2 / HNBR 3 - RDB: 0.5% at 180 min — Table 2 / HNBR 1 - RDB: 3.9% at 180 min — Table 2 / HNBR 2 - RDB: 1.3% at 180 min — Table 2 / HNBR 3 - RDB: 0.4% at 240 min — Table 2 / HNBR 1 - RDB: 3.5% at 240 min — Table 2 / HNBR 2 - RDB: 0.7% at 240 min — Table 2 / HNBR 3 - Mw: 113371 g/mol — Table 3 / HNBR 1 - Mn: 48352 g/mol — Table 3 / HNBR 1 - PDI: 2.34 — Table 3 / HNBR 1 - Mooney viscosity: 33.3 — Table 3 / HNBR 1 - Mw: 133758 g/mol — Table 3 / HNBR 2 - Mn: 54595 g/mol — Table 3 / HNBR 2 - PDI: 2.45 — Table 3 / HNBR 2 - Mooney viscosity: 41.3 — Table 3 / HNBR 2 - Mw: 195304 g/mol — Table 3 / HNBR 3 - Mn: 64130 g/mol — Table 3 / HNBR 3 - PDI: 3.05 — Table 3 / HNBR 3 - Mooney viscosity: 69.3 — Table 3 / HNBR 3

Relevant Experimental Evidence - Example 1: Hoveyda-type catalyst without co-catalyst gave 2.5% RDB at 60 min and 0.4% at 240 min. - Example 1: Arlt catalyst without co-catalyst gave slower early hydrogenation than Hoveyda. - Example 1: Wilkinson's catalyst with triphenylphosphine showed the slowest hydrogenation profile. - Example 1: HNBR 1 had lower Mw and Mooney viscosity than the industrial-norm HNBR 3. - Example 1: The catalyst system improved hydrogenation efficiency while reducing molecular weight.

Technical Relevance This patent is relevant because it provides a catalyst-system comparison for simultaneous hydrogenation/metathesis routes to low-molecular-weight HNBR. It shows that catalyst structure and co-catalyst selection can materially improve hydrogenation rate and product rheology.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US10508156B2 emphasizes catalyst pretreatment with vinyl ether co-catalysts to improve hydrogenation activity while preserving molecular weight, whereas EP2473280B1 and US9150669B2 focus on simultaneous metathesis/hydrogenation to intentionally lower molecular weight and Mooney viscosity.
- US10508156B2 uses solution hydrogenation at 120°C and 4.137 MPa in the main examples, while EP2473280B1 and US9150669B2 use 140°C and 8.4 MPa in their comparative hydrogenation profiles.
- EP3827029A1 is distinct in that it centers on noble-metal loading windows and catalyst residue/aging control rather than metathesis-driven molecular-weight reduction.
- Across the metathesis-based patents, ruthenium/osmium catalysts are the dominant platform; EP2473280B1 explicitly compares Grubbs III, Grubbs II, and Wilkinson's catalyst, while US9150669B2 compares Hoveyda, Arlt, and Wilkinson-type systems.
- US7772328B2 and EP2473280B1 both disclose co-olefin-assisted chain scission before or during hydrogenation, with ethylene and 1-hexene used to reduce Mw and PDI.
- US10508156B2 shows that vinyl ethyl ether pretreatment can accelerate hydrogenation and reduce PDI relative to untreated catalyst systems, supporting catalyst activation as a key process lever.
- EP3827029A1 specifies broad but explicit catalyst concentration ranges: Ru 10-200 ppm, Pd 20-<67 ppm, Rh 50-<270 ppm, with hydrogenation at 60-200°C for 1-24 h.
- The patents collectively indicate that low residual unsaturation, controlled catalyst loading, and careful management of metathesis side activity are central to producing processable HNBR with acceptable molecular weight and aging stability.

4. CONCLUSION

The evidence shows two viable HNBR synthesis paradigms: direct selective hydrogenation of NBR with optimized noble-metal catalysts, and one-step metathesis/hydrogenation to deliberately lower molecular weight and Mooney viscosity. For high-performance HNBR, the strongest evidence supports solution-phase processing in organic solvent with hydrogenation temperatures around 120-140°C, hydrogen pressures around 4.137-8.4 MPa, and catalyst systems based on Ru/Os or related noble-metal complexes. Catalyst pretreatment with vinyl ethers or use of carefully chosen co-catalysts can improve hydrogenation rate and help preserve molecular weight, while co-olefins such as ethylene or 1-hexene can be used when molecular-weight reduction is desired. EP3827029A1 further indicates that catalyst residue control is important for aging stability and product appearance. Based on the disclosed evidence, recommended synthesis parameters are: solution hydrogenation, Ru/Os-based catalyst systems, hydrogen pressure in the 4-8.4 MPa range, temperature near 120-140°C, and explicit control of catalyst loading and pretreatment to balance hydrogenation efficiency, residual unsaturation, and final rheology.

5. REFERENCES

- US10508156B2 | Catalyst compositions and their use for hydrogenation of nitrile rubber | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10508156B2/en>
- US7772328B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7772328B2/en>
- EP3827029A1 | Method for producing hydrogenated nitrile rubber and hnbr compositions thereof | Not disclosed | EP | 2021 | <https://patents.google.com/patent/EP3827029A1/en>
- EP2473280B1 | Process for the preparation of hydrogenated nitrile rubber | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473280B1/en>
- US9150669B2 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9150669B2/en>